Abhinay Kumar Sahu

sahuabhinay317@gmail.com · +91-77228-67533 · Bhilai, Chhattisgarh · LinkedIn · GitHub

SUMMARY

BCA undergraduate at Christ University with hands-on experience in Android development, full-stack web development, and backend systems. Skilled in building offline-first mobile applications, designing component-driven web interfaces, and applying core computer science principles to real-world problems. Experienced in delivering complete, user-focused projects that combine clean architecture with practical software engineering.

EDUCATION

Bachelor of Computer Applications (BCA) — Christ University *2024 – 2028*
Pune Lavasa Campus · Coursework: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Class XII — Delhi Public School Risali *Completed 2024*
Bhilai, Chhattisgarh

TECHNICAL SKILLS

Languages	C++, Kotlin, Java, JavaScript (ES6+), SQL
Android Dev	Android Studio, Kotlin, XML, Room DB, SQLite, Material Design 3, Jetpack Navigation, MVVM, LiveData, ViewModel, Kotlin Coroutines
Web Dev	React.js, HTML5, CSS3
Backend & DBs	Firebase (Auth, Firestore), MySQL, RESTful APIs
Core CS	OOP, Data Structures & Algorithms, Memory Management, Design Patterns
Tools	Git, GitHub, Android Studio, VS Code, Figma, Postman

PROJECTS

Flash Deck — Offline Study Assistant *2025*
Android Studio • Kotlin • Room Database • SQLite

- Native Android flashcard app with an integrated AI chatbot; supports unlimited decks with spaced-repetition scheduling, reducing average study session length by an estimated 30%
- Full offline functionality via Room Database and SQLite, eliminating internet dependency for 100% of core features and enabling use in zero-connectivity environments
- MVVM architecture with LiveData and ViewModel; Kotlin coroutines keep chatbot responses off the main thread, maintaining 60fps UI performance
- Material Design 3 UI with custom card-flip animations; tested and iterated with 8 users based on structured feedback

Lavasa Travel Guide — Tourism Web Application *2025*
React.js • JavaScript • CSS3 • HTML5

- Single-page application with reusable React components for attractions, accommodations, and activity listings
- React state management for dynamic tour planning, reducing page reloads to zero and improving content interaction speed over a static equivalent
- Fully responsive layout across desktop and mobile using CSS Flexbox and Grid, with smooth transitions and sub-100ms interaction feedback

Watch & SmartWatch Inventory System *2024*
C++ • OOP • STL

- Class hierarchy with `Watch` base and `SmartWatch` derived class, demonstrating inheritance, polymorphism, and encapsulation
- `std::unique_ptr` for memory safety; `dynamic_cast` for runtime type identification on derived objects
- CLI with full CRUD operations via virtual dispatch (`showDetails()`, `getType()`) and virtual destructor for correct cleanup

Interactive Ping Pong Game *2024*

HTML5 Canvas • CSS3 • JavaScript

- Browser-based 2-player game with Canvas API rendering at stable 60fps, velocity-based ball physics, and per-frame collision detection
- OOP design for paddle and ball entities; keyboard-driven game-state management with score tracking and round resets

Multi-Screen Navigation Application *2024*

Android Studio • Kotlin • Jetpack Navigation

- Fragment-based navigation across 3 screens with back stack management, deep linking support, and Material Design components

ACHIEVEMENTS & LEADERSHIP

Hackverse 2024 — Qualified for Round 2 in a competitive college-level hackathon; built a working prototype under a strict time constraint

Student Coordinator, Gaming Club — Organized 10+ weekly game-dev and design sessions; managed club events and member engagement activities

Gaming Competition Representative — Represented Christ University in an inter-college gaming tournament